Balram Maurya

Patiala, Punjab, India | +91-6284595800 | balrammaurya2005@gmail.com | linkedin.com/in/balram-maurya-639026273

Professional Summary

Hands-on electronics and embedded-systems student with experience building connected robotics, computer-vision-assisted monitoring systems, and drone prototypes. Skilled in ESP32-based integration, sensor interfacing, motor and power integration, wireless communication, debugging, and iterative prototyping. Motivated to apply practical engineering skills to safe, reliable, and accessible electric mobility products, with strong interest in power electronics and EV systems.

Education

Thapar Institute of Engineering & Technology, Patiala *Expected May 2027*
Bachelor of Engineering in Electronics & Communication Engineering

Technical Skills

Embedded Systems: ESP32, Arduino, Embedded C/C++, sensor interfacing, I2S, Wi-Fi communication, firmware debugging
Electronics & Power: ESC-based brushless motor control, DC motor drive, Li-Po battery integration, DC-DC buck regulation, power distribution and wiring, OLED and audio peripherals
Sensing & Control: MPU6050/IMU interfacing, sensor data acquisition, calibration, motion-sensing experimentation
Software & AI: Python, OpenCV, MediaPipe, Random Forest, AI/LLM API integration
Engineering Practices: Requirements, prototyping, bench testing, troubleshooting, hardware-software integration, documentation

Projects

NEO — AI-Powered Desktop Companion & Mobile Robot *Prototype (near completion)*

- Developed NEO, an AI-enabled desktop companion that monitors user attention and fatigue via real-time eye tracking and blink analysis (Python, OpenCV, MediaPipe), classifying cognitive load into Low, Medium, and High states with a Random Forest pipeline to trigger rest reminders.
- Integrated ESP32 with an SSD1306 OLED, INMP441 I2S microphone, and MAX98357A audio module in C++ firmware to drive animated facial expressions and voice interaction on one embedded platform.
- Designed and assembled a four-wheel mobile robot with motor control, battery integration, and on-board power regulation delivering stable supply rails to logic, audio, and drive subsystems.
- Implemented Wi-Fi communication between the desktop application and ESP32 for wireless delivery of commands and AI responses, resolving connectivity, timing, and power faults through system-level debugging.

Autonomous Drone System — ISRO Robotics Challenge (IRoC-U 2025) *Team AGNIV*

- Selected for Round 2 of IRoC-U 2025 as a member of Team AGNIV, contributing to the design and planning of an autonomous drone navigation solution.
- Evaluated system architecture and components against power, payload, and reliability requirements, and prepared supporting technical documentation.
- Contributed to sensor integration, data acquisition, and system testing across Pixhawk, Raspberry Pi, LiDAR, and IMU subsystems within a multidisciplinary engineering team.

ESP32-Based Quadcopter Prototype *First-Year Hardware Project*

- Built an ESP32-based quadcopter prototype, integrating an MPU6050 inertial measurement unit with four motor-control channels for real-time motion sensing and flight-control experimentation.
- Wired brushless DC motors, electronic speed controllers, Li-Po power distribution, and a DC-DC buck regulator for a stable low-voltage electronics supply, building practical embedded power-integration skills.
- Implemented and tested sensor-data acquisition and basic motor-response logic through iterative prototyping, hardware calibration, and connection-level debugging.

Leadership Experience

Finance Secretary — Institution's Innovation Council (IIC), Thapar Institute *2025 – 2026*

- Managed a budget of approximately INR 2L+ across 8+ institutional events through resource allocation, expense tracking, documentation, and financial reporting.
- Coordinated with faculty, student teams, and external stakeholders to support planning, procurement, and operational execution across multiple initiatives.
- Developed structured Excel-based budgeting and reporting workflows to improve documentation consistency and resource planning.

Innovation Ambassador — Institution's Innovation Council (IIC), Ministry of Education | Participant, Axcelrate 2026

Additional Experience

Author (Pen Name: Sterling Moriz) *Amazon Kindle*

- Published 5+ eBooks under the pen name Sterling Moriz, including *5 Proven Secrets to Crush Procrastination* and *Master AI Productivity*, co-authored with Alex Ross.
- Managed publications end to end, from topic research and planning through writing, editing, digital publishing, and market positioning.